UNIVERSITY INTERSCHOLASTIC LEAGUE

Mathematics

Invitational B • 2026

DO NOT TURN THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO!

1. The Mean Green Gym in Denton charges members \$30.00 per month, plus \$1.50 every time a member comes in to exercise. They also charge \$1.00 if you need a towel. In a typical month, Ted exercises 26 times and he needs a towel 13 times. What is Ted's annual cost to exercise at the gym?

(A) \$976.00 (B) \$978.00 (C) \$980.00 (D) \$982.00 (E) \$984.00

2. Emma starts at Rankin High School and walks 12 blocks north. Then she turns and walks 18 blocks west. Next, she turns and walks 22 blocks south and stops for lunch at The Red Devil Pizzeria. If each block in Rankin is one-eighth of a mile long, how far is The Red Devil Pizzeria from the high school? (nearest hundredth)

(A) 2.54 mi (B) 2.57 mi (C) 2.60 mi (D) 2.63 mi (E) 2.66 mi

3-4. Consider ΔABC with $m\angle A = 6x,\ m\angle B = 8x+2, m\angle C = 12x-4,$ and $AB = 17.$

3. ΔABC is a/an ______________ triangle.

(A) acute (B) right (C) obtuse (D) isosceles (E) equilateral

4. The perimeter of ΔABC is ___________. (nearest tenth)

(A) 43.0 (B) 43.2 (C) 43.4 (D) 43.6 (E) 43.8

5. Consider three consecutive Fibonacci numbers such that the product of the numbers is 166,430. Find the sum of the three numbers.

(A) 176 (B) 177 (C) 178 (D) 179 (E) 180

6. Dr. Roach plans to hire Adrian, Jeffry, Caleb, and Aidyn to paint the 24 classrooms at the new elementary school in Olney during the summer. Adrian can paint a classroom in 10 hours, Jeffry can paint a classroom in 9 hours, Caleb can paint a classroom in 12 hours and Aidyn can paint a classroom in 15 hours. If they work together, how much time is required to paint the 24 classrooms? (nearest minute)

(A) 66 hr 22 min (B) 66 hr 25 min (C) 66 hr 28 min (D) 66 hr 31 min (E) 66 hr 34 min

7. James is taking Modern Physics this semester at MIT (a long way from Knippa). His lab average is 35% of his grade, his test average is 45% of his grade and his semester exam score is 20% of his grade. James has a 93 lab average. His test scores are 87, 91, 84 and 92. An overall average of 90.00 or higher earns an A for the semester. What is the minimum semester exam score needed to earn an A?

(A) 87 (B) 88 (C) 89 (D) 90 (E) 91

8. If $\dfrac{x+3}{x^2-7x+12} - \dfrac{x-4}{x^2-9} = \dfrac{ax^2+bx+c}{dx^4+ex^3+fx^2+gx+h}$, then $\dfrac{a+b+c}{d+e+f+g+h} =$ ________.

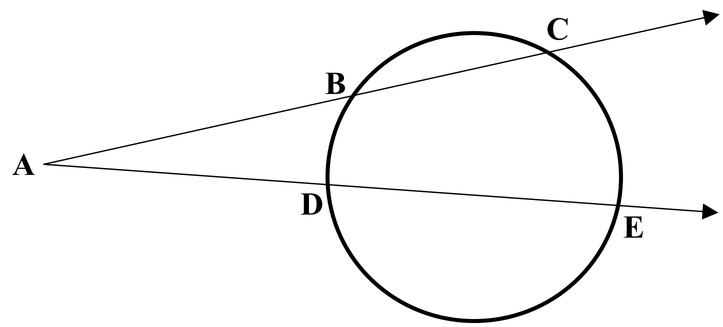
(A) $\frac{5}{24}$ (B) $\frac{1}{4}$ (C) $\frac{7}{24}$ (D) $\frac{1}{3}$ (E) $\frac{3}{8}$

9. A right circular cone has a diameter of 15 and a slant height of 11. Find the volume of the cone. (nearest whole number)

(A) 472 (B) 474 (C) 476 (D) 478 (E) 480

10-11. Consider the circle shown with minor arcs CE and BD.

Given: $mCE = 82°$, $mBD = 50°$, AB = 24, BC = 16, and AD = 20.



Problems 10, 11

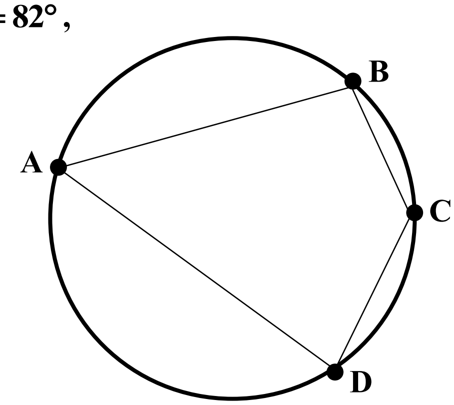
10. $m\angle CAE =$ ________. (nearest tenth)

(A) 16.0°
(B) 16.2°
(C) 16.4°
(D) 16.6°
(E) 16.8°

11. DE = ________. (nearest tenth)

(A) 27.2 (B) 27.4 (C) 27.6 (D) 27.8 (E) 28.0

12-13. Consider the circle shown with $m\angle DAB = 60°$, $m\angle ADC = 82°$, and minor arc BC with $mBC = 52°$.



Problems 12, 13

12. $m\angle BCD =$ _______.

(A) 116°
(B) 117°
(C) 118°
(D) 119°
(E) 120°

13. The measure of minor arc AD is _______. (nearest whole number)

(A) 124° (B) 126° (C) 128° (D) 130° (E) 132°

14. If $f(x) = x^2 + 8x + 16$ and $g(x) = \sqrt{x+23}$, then $(f \circ g)(2) =$ ________.

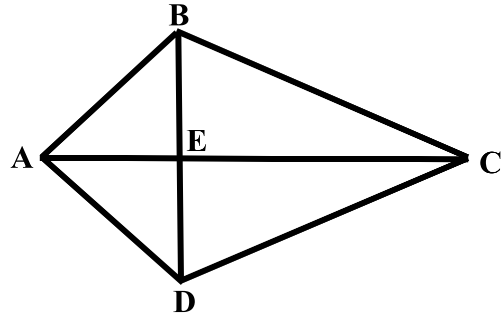
(A) 72 (B) 75 (C) 78 (D) 81 (E) 84

15. Let $g(x)$ be the inverse function of $f(x) = 4x^2\sqrt{x}$. Find the smallest positive integer value of x such that $g(x) > 2$.

(A) 21 (B) 22 (C) 23 (D) 24 (E) 25

16. Kite ABCD has an area of 216.
If BE = 9, then AC = _______. (nearest whole number)

(A) 22
(B) 23
(C) 24
(D) 25
(E) 26

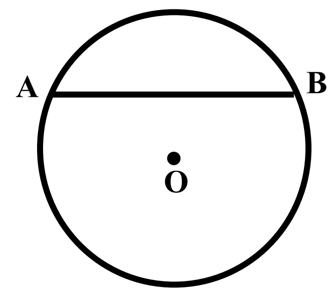


Problems 16, 17

17. If $AB = \sqrt{145}$, then $m\angle BCE =$ _________. (nearest tenth)

(A) 28.5° (B) 28.8° (C) 29.1° (D) 29.4° (E) 29.7°

18-19. Consider the circle with center O shown on the right. The length of chord $\overline{AB}$ is 44. The area of the circle is 1901.



Problems 18, 19

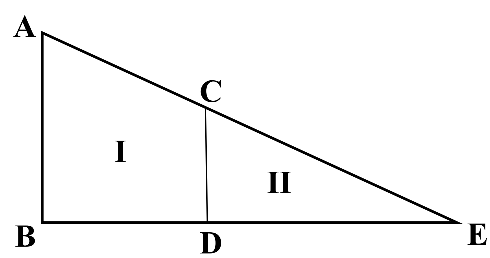
18. Find the arclength of minor arc $\overset{\frown}{AB}$. (nearest tenth)

(A) 53.6
(B) 53.9
(C) 54.2
(D) 54.5
(E) 54.8

19. Find the area bounded by minor arc $\overset{\frown}{AB}$ and chord $\overline{AB}$. (nearest whole number)

(A) 416 (B) 419 (C) 422 (D) 425 (E) 428

20-21. $\overline{AB}$ is parallel to $\overline{CD}$ and $\overline{CD} \perp \overline{BE}$.
$AB = 12,\ CD = 8,\ CE = 21.54$



Problems 20, 21

20. AC = _________. (nearest hundredth)

(A) 10.77 (B) 10.88 (C) 10.99
(D) 11.10 (E) 11.21

21. Find the area of region I. (nearest whole number)

(A) 98 (B) 100 (C) 102 (D) 104 (E) 106

22. If s(x) is the slant asymptote of the graph of $h(x) = \dfrac{3x^2 - 6x + 11}{x + 2}$, then h(48) – s(48) = _______.
(nearest hundredth)

(A) 0.58 (B) 0.61 (C) 0.64 (D) 0.67 (E) 0.70

23. Coach Perkins has 2 posts, 8 wings and 6 guards on his team. His starting lineup must consist of one post, two wings and two guards. How many starting lineups are possible?

(A) 828 (B) 832 (C) 836 (D) 840 (E) 844

24. When the vector $v = 12i - 9j$ is converted to polar coordinates, one correct answer is $v = \langle 15, \theta° \rangle$. θ could be _________°. (nearest whole number)

(A) 317 (B) 319 (C) 321 (D) 323 (E) 325

25-26. The length of a side of a regular dodecagon is 18. The dodecagon is inscribed in a circle.

25. The area of the dodecagon is __________. (nearest whole number)

(A) 3616 (B) 3619 (C) 3622 (D) 3625 (E) 3628

26. The area of the circle that the dodecagon is inscribed in is ________. (nearest whole number)

(A) 3796 (B) 3799 (C) 3802 (D) 3805 (E) 3808

27. The equation of the line that is equidistant from the points $(-7, 8)$ and $(9, -4)$ is $ax - 3y + c = 0$. $a + c =$ __________.

(A) 3 (B) 4 (C) 5 (D) 6 (E) 7

28. Consider a geometric sequence in which the 2^{nd} term is 25.6 and the 5^{th} term is 13.1072. Find the sum of the first 12 terms of the sequence. (nearest thousandth)

(A) 148.997 (B) 149.001 (C) 149.005 (D) 149.009 (E) 149.013

29. Cindy is flying a kite. The kite string has a length of 360 ft and the angle of elevation from Cindy to the kite is 48.6°. If the end of the string Cindy is holding is 4 ft above the ground, how high above the ground is the kite? (nearest foot)

(A) 274 ft (B) 276 ft (C) 278 ft (D) 280 ft (E) 282 ft

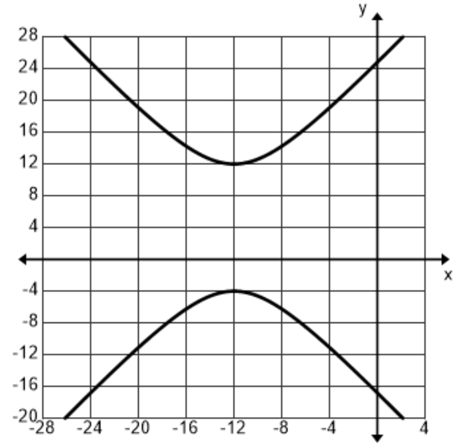
30. Four statistics books and seven calculus books are on a shelf in my study. How many ways can they be arranged if the statistics books are kept together and the calculus books are kept together?

(A) 60,480 (B) 120,960 (C) 181,440 (D) 241,920 (E) 302,400

31. My clock shows that it is exactly 7:45. How long will I have to wait until both hands (hour hand and minute hand) of my clock point in the same direction for the second time? (nearest second)

(A) 124 min 5 sec (B) 124 min 9 sec (C) 124 min 13 sec (D) 124 min 17 sec (E) 124 min 21 sec

32-34. The point $P(0, 24.8)$ lies on the hyperbola.



Problems 32, 33, 34

32. The distance between the foci is _________. (nearest tenth)

(A) 18.5
(B) 18.7
(C) 18.9
(D) 19.1
(E) 19.3

33. The equation of the asymptotes of the hyperbola are $y_1 = ax + b$ and $y_2 = cx + d$. $b + d =$ _______. (nearest tenth)

(A) 7.6 (B) 7.8 (C) 8.0 (D) 8.2 (E) 8.4

34. The parametric equations that produce the graph of the hyperbola are $x(t) = h\tan(t) + i$ and $y(t) = k\sec(t) + j$ for $-2\pi \le t \le 2\pi$. $h + i + j + k =$ _________.

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

35. The half-life of carbon-14 is thought to be 5730 years. In 1988, a laboratory in Zurich examined the Shroud of Turin and dated the shroud to 1312. What percentage of the original amount of carbon-14 remained in 2026? (nearest tenth)

(A) 91.3% (B) 91.5% (C) 91.7% (D) 91.9% (E) 92.1%

36. By road, it is 653 miles down I-80 from York, Nebraska to Rock Springs, Wyoming. Joe leaves York at 8:15 AM (central time) and heads west toward Rock Springs at an average speed of 74 mph. Suzy leaves Rock Springs at 9:30 AM (central time) and heads east toward York at an average speed of 68 mph. How far are they from York when they meet? (nearest whole number)

(A) 383 mi (B) 385 mi (C) 387 mi (D) 389 mi (E) 391 mi

37. Given: $f^{-1}(x) = \dfrac{ax - 5}{-3x + 2}$ and $f^{-1}(1) = 9$. Evaluate $f(2)$. (nearest tenth)

(A) 4.2 (B) 4.5 (C) 4.8 (D) 5.1 (E) 5.4

38. When you get into a Ferris Wheel car at the bottom of the ride, you are 2 feet off the ground. When the ride starts and the Ferris Wheel begins rotating, it takes 12 seconds to reach the highest point of the ride where you are 72 feet off the ground. Ella gets into a car at the bottom and the Ferris Wheel rotates for 39 seconds and stops. How far above the ground is Ella? (nearest inch)

(A) 60 ft 9 in (B) 61 ft 1 in (C) 61 ft 5 in (D) 61 ft 9 in (E) 62 ft 1 in

39-40. Given: $y_1 = -0.25(x+6)^2 - 2$ and $y_2 = 0.5x - 4$

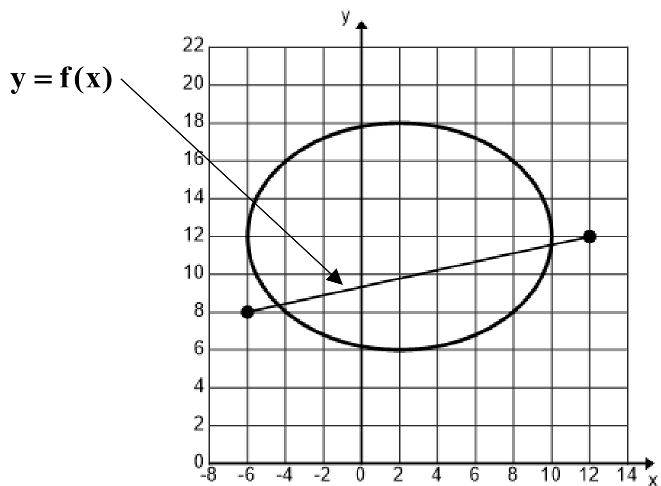
39. The point $F(a, b)$ is the focus of the graph of y_1. $a+b =$ __________.

(A) -12 (B) -10 (C) -9 (D) -8.5 (E) -8.25

40. The graphs of the two curves intersect at points P and R. Find the distance from P to R. (nearest tenth)

(A) 9.4 (B) 9.6 (C) 9.8 (D) 10.0 (E) 10.2

41-45. Given the graph of the ellipse with vertices $(-6, 12)$ and $(10, 12)$ and containing the points $P(2, 18)$ and $Q(8, b), b < 9$. The line $y = f(x)$ intersects the ellipse at points C and D.



Problems 41, 42, 43, 44, 45

41. Find the slope of the line tangent to the graph of the ellipse at point Q. (nearest hundredth)

(A) 0.63
(B) 0.74
(C) 0.85
(D) 0.96
(E) 1.07

42. Find the area of the region bounded by the graphs of $y = f(x)$ and minor arc CD. (nearest whole number)

(A) 42 (B) 43 (C) 44 (D) 45 (E) 46

43. The region bounded by the graphs of $y = f(x)$ and minor arc CD is revolved about the line $y_2 = -2$. Find the volume of the solid created. (nearest whole number)

(A) 2747 (B) 2750 (C) 2753 (D) 2756 (E) 2759

44. Find the volume of a solid using cross sections perpendicular to the x-axis if the base of the solid is the region bounded by the graphs of $y = f(x)$ and minor arc CD. Each cross section is a square. (nearest whole number)

(A) 137 (B) 140 (C) 143 (D) 146 (E) 149

45. Find the arclength of minor arc CD. (nearest tenth)

(A) 16.4 (B) 16.7 (C) 17.0 (D) 17.3 (E) 17.6

46. The first floor of the Montana Tech Sports Performance Center consists of a rectangle with a semicircle on each end. A 300-meter track runs around the outside. If the designers of the indoor track wanted to maximize the rectangular area, the radius of each semicircle should be __________ meters. (nearest tenth)

(A) 23.3 m (B) 23.5 m (C) 23.7 m (D) 23.9 m (E) 24.1 m

47. Newton's Law of Cooling states that the rate of change in the temperature of an object is proportional to the difference between the object's temperature and the temperature of the surrounding medium. Consider an object placed in a room kept at a constant temperature of 65°. At t = 0, the temperature of the object is 190°. At t = 30 min, the temperature of the object is 125°. The temperature of the object should reach 70° at t = __________. (nearest whole number)

(A) 120 min (B) 123 min (C) 126 min (D) 129 min (E) 132 min

48-50. A particle is traveling along the x-axis. At t = 0, the position of the particle is at x = 2 cm. The velocity of the particle is given in cm/s by $v(t) = 0.5t^3 - 6t^2 - 4t - 9,\ 0 \le t \le 14$.

48. Find the maximum speed of the particle. (nearest whole number)

(A) 130 cm/s (B) 140 cm/s (C) 150 cm/s (D) 160 cm/s (E) 170 cm/s

49. The position of the particle when the acceleration of the particle is 112 cm/s^2 to the right is at x = ____________ cm. (nearest negative integer)

(A) −1240 (B) −1232 (C) −1224 (D) −1216 (E) −1208

50. Find the total distance traveled by the particle on the time interval $[0, 14]$. (nearest whole number)

(A) 1344 cm (B) 1348 cm (C) 1352 cm (D) 1356 cm (E) 1360 cm

51. An airplane flies directly over a stationary observer. The plane is flying horizontally at a constant speed of 400 mph at an altitude of 1.50 miles. The rate at which the angle of elevation from the observer to the plane is decreasing when the plane is a horizontal distance of 2.00 miles from the observer is _________ rad/hr. (nearest whole number)

(A) 92 (B) 96 (C) 100 (D) 104 (E) 108

52. Determine the interval of convergence for the power series $\sum_{n=1}^{\infty} \frac{(-1)^n n(x+5)^n}{3^n}$.

(A) $(-8, -2]$ (B) $[-8, -2)$ (C) $(-8, -2)$ (D) $[-8, -2]$ (E) $(-\infty, \infty)$

53. Suppose the distribution of the length of drives by Rory McIlroy in his career is approximately normal with a mean of 320 yards and a standard deviation of 15 yards. What percent of his drives are between 335 yards and 350 yards? (nearest tenth)

(A) 12.5% (B) 13.6% (C) 14.7% (D) 15.8% (E) 16.9%

Advanced Math Classes	2	3	4	5	6
Probability	.12	.24	.32	.20	.12

54-55. Students at SEM are required to take at least two advanced math classes during their four high school years. They are not allowed to take more than six advanced math classes. Define X to be the number of advanced math classes that a randomly selected student takes.

54. Find the mean of the random variable X. (nearest hundredth)

(A) 3.94 (B) 3.96 (C) 3.98 (D) 4.00 (E) 4.02

55. Find the standard deviation of the random variable X. (nearest hundredth)

(A) 1.12 (B) 1.14 (C) 1.16 (D) 1.18 (E) 1.20

56. Stephen took Abstract Algebra from Dr. Pony at SMU his senior year. He scored a 77 on the semester exam, which concerned him at first. But when Dr. Pony revealed that the mean score on the exam was 56, with a standard deviation of SD, he felt better because his score placed him at the 97^{th} percentile. SD = _________. (nearest whole number)

(A) 9 (B) 10 (C) 11 (D) 12 (E) 13

Day	Sun	Mon	Tues	Wed	Thur	Fri	Sat
# C.E.	68	52	48	52	n	76	63

57-60. The number of patients who came to the emergency room at the Mount Angel Medical Center in 2024 is shown in the table. Jeannie analyzed the data and performed a Chi-Square GOF Test with the following null hypothesis.
H_0: ER visits at the Medical Center are evenly distributed across the days of the week.

57. If the mean number of ER visits for each day is 60.0, what is the value of n?

(A) 58 (B) 59 (C) 60 (D) 61 (E) 62

58. The IQR of the data is __________. (nearest whole number)

(A) 13 (B) 14 (C) 15 (D) 16 (E) 17

59. The Chi-Square statistic is ________. (nearest tenth)

(A) 10.0 (B) 10.2 (C) 10.4 (D) 10.6 (E) 10.8

60. The Friday cell contributed ________ to the Chi-Square statistic. (nearest tenth)

(A) 3.5 (B) 3.7 (C) 3.9 (D) 4.1 (E) 4.3

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University Interscholastic League
MATHEMATICS CONTEST
HS ● Invitational B ● 2026
Answer Key

1. E	21. B	41. C
2. B	22. E	42. A
3. A	23. D	43. C
4. B	24. D	44. B
5. C	25. E	45. E
6. C	26. B	46. D
7. C	27. D	47. E
8. C	28. C	48. E
9. B	29. A	49. A
10. A	30. D	50. E
11. E	31. A	51. B
12. E	32. C	52. C
13. C	33. C	53. B
14. D	34. A	54. B
15. C	35. C	55. D
16. C	36. B	56. C
17. D	37. B	57. D
18. D	38. D	58. D
19. E	39. C	59. A
20. A	40. E	60. E